

Section: **3.4 – STRENGTH OF COMPOSITE PARTS**

Chapter: **3465**

Title: **HOLED PLATES AND WEBS**

Revision: **0.3**

Date: **29/11/2018**

1 INTRODUCTION

1.1 OBJECT

This chapter describes the methodology proposed for calculating the buckling onset and the strength of holed rectangular plane plates or composite laminates.

Methods presented are approximated, to be used in initial phases of the design process. They are based in solutions or approximations found in literature, engineering handbooks, typically for isotropic materials, and “extrapolated” to composite laminates.

For final sizings or certification strength justifications, it is strongly recommended that numerical FEM analysis techniques are used (and even dedicated tests support in case exploitation of post-buckling material capabilities is need to reach positive safety margins).

1.2 SCOPE AND LIMITATIONS

Strength of composite rectangular plates with circular holes.

Pre-sizing methods presented in this chapter are rough and approximated. They shall be only used in initial phases of the design process.

2 METHODS FOR PRE-SIZING

2.1 STRESS AND STRENGTH ANALYSIS

Stress field in the infinite plate

ESDU85001 sheet presents the solution of the stress field around a hole in an infinite plate made of an orthotropic fibrous composite laminate.

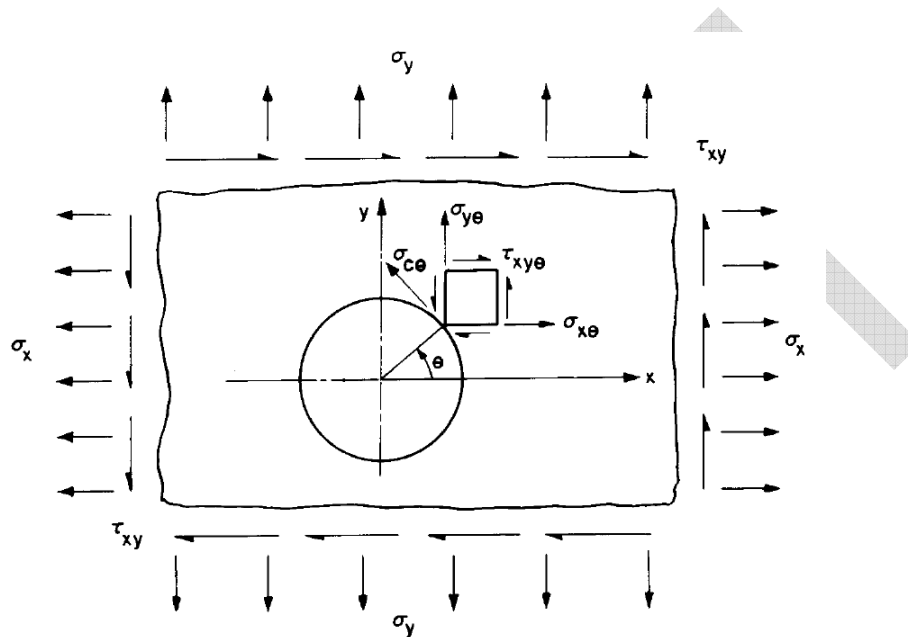


Figure 1. Circular hole in infinite plate

The following step shall be to correct the stresses/strains obtained with that method to account for the limited plate width (in case this effect is applicable, i.e. when plate ends at supports).

Bending loads and non-symmetric laminates

In case the laminate plate is subjected to out-plane bending or if the laminate is un-symmetric/un-balanced and thus, the shear-extensional coupling exists (the [B] stiffness matrix not zero), the following approach is suggested...

1. Calculate the mid-plane strain in the un-holed laminate using the classical laminate theory (see chapter §3401 "STRESS AND STRAIN IN COMPOSITE LAMINATES"). Using in-line notation for the plate loads, laminate stiffness matrices and mid-plane strains, it becomes:

$$(N_x, N_y, N_{xy}, M_x, M_y, M_{xy}) = [ABBD] \cdot (\epsilon_x, \epsilon_y, \gamma_{xy}, \kappa_x, \kappa_y, \kappa_{xy})$$

2. Calculate, from the laminate mid-plane strains, the strains at the upper and lower faces of the laminate as:

$$(\epsilon_x, \epsilon_y, \gamma_{xy})_{\text{upper}} = (\epsilon_x, \epsilon_y, \gamma_{xy}) + (t/2) \cdot (\kappa_x, \kappa_y, \kappa_{xy})$$

$$(\epsilon_x, \epsilon_y, \gamma_{xy})_{\text{lower}} = (\epsilon_x, \epsilon_y, \gamma_{xy}) - (t/2) \cdot (\kappa_x, \kappa_y, \kappa_{xy})$$

3. Get the equivalent in-plane loads at each of the faces that would correspond to an orthotropic laminate with the same in-plane stiffness:

$$(N_x, N_y, N_{xy})_{upper} = [A] \cdot (\epsilon_x, \epsilon_y, \gamma_{xy})_{upper}$$

$$(N_x, N_y, N_{xy})_{lower} = [A] \cdot (\epsilon_x, \epsilon_y, \gamma_{xy})_{lower}$$

4. Use the abovementioned ESDU85001 methodology to get the stress/strain distribution around the hole edge at upper and at lower faces, using the same [A] in-plane stiffness in the analysis.

Warning! Previous approximation/approach has not been checked nor validated. A linear strain distribution through thickness could be expected to be found at hole edge. However, the stiffness distribution is not constant across and this effect, accounted in un-holed plates by the [D] matrix, has been encompassed in this stress concentration analysis. On the other hand, it is known from isotropic materials that the typical stress concentration factor for plate bending (typ. 2) is lower than for axial load (typ. 3). **ToDo. Cross-check with FEM simulations.**

Finite Width Correction

- ✓ Axial loads (X direction)

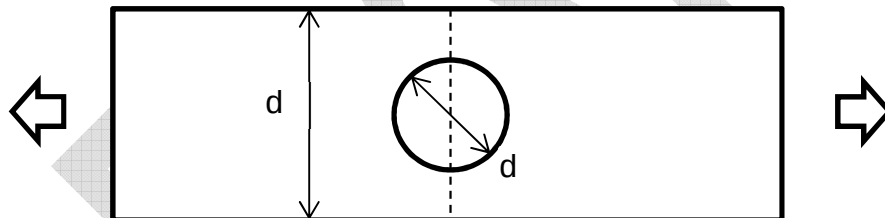


Figure 2. Notched plate with finite width

The following relationship between the stress concentration factors for the infinite plate and a finite width strip, proposed by Tan (expression for circular holes on QI laminates), shall be applied to the stress/strain field around hole edge obtained for the infinite plate in pure axial loads...

$$K_T / K_{T0} = [2 + (1 - d/b)^3] / [3 \cdot (1 - d/b)] \quad (\text{section cut normal to load})$$

- ✓ Shear loads

The ESDU75034 sheets propose a stress correction for shear webs in isotropic material. Referred to the reference value of the stress concentration factors of infinite plates in in-plane shear, can be applied to the stress/strain field around hole edge obtained for the infinite plate in pure shear...

$$K_T / K_{T0} = 1 + 9 \cdot (d/b)^3 \quad (\text{section cuts at } \pm 45^\circ)$$

See following section 2.2. The proposed approximation for f_{max}/q done by the ESDU was divided by the typical stress concentration factor for shear in infinite plate, i.e. by 4.

Strength check

Once the stress distribution around the hole is known, the strength check and resulting safety margin shall be obtained according to method described in chapter 3404 "STRENGTH IN NOTCHED LAMINATES" (note that a straight forward comparison of the peak stress at hole edge and the material strength shall be overly conservative).

If the holed plate has damage tolerance requirements, and strength including BVID needs to be demonstrated, dedicated test supported material allowable strengths shall have to be available (at present time no material models are available to anticipate residual TAI & CAI strengths at parts locations with stress raisers as holes).

2.2 BUCKLING STABILITY

The ESDU75034 sheet has been typically used to correct analyses on metallic shear webs (rectangular plates of isotropic materials) to account the effect of centred circular holes.

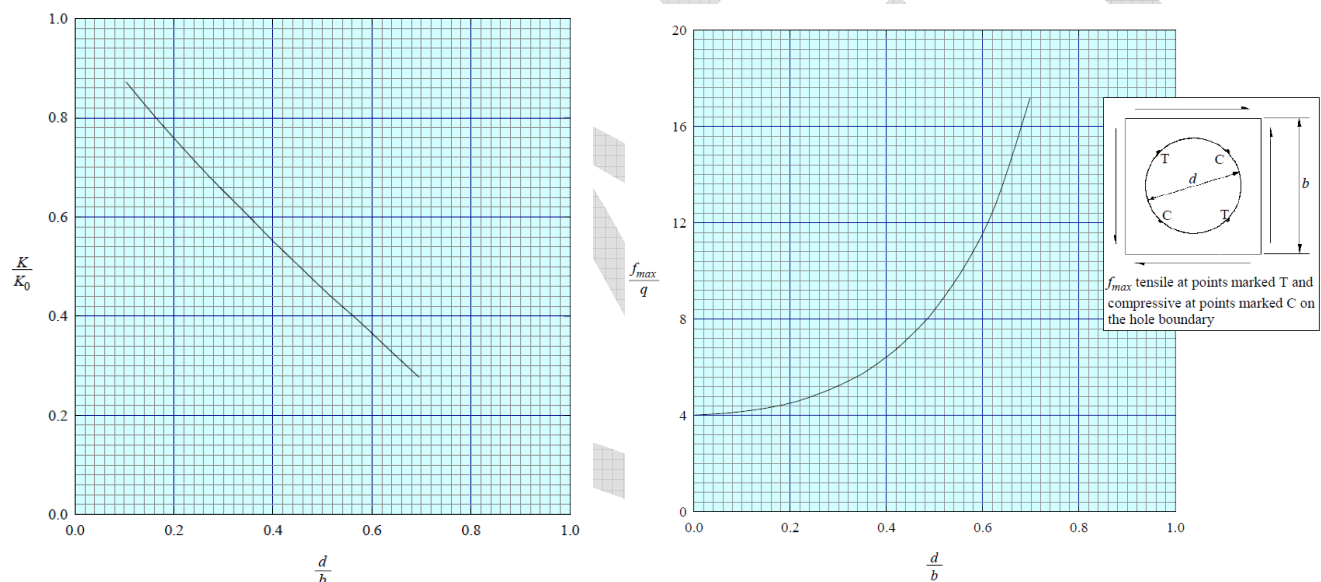


Figure 3. Shear buckling correction and stress concentration due to hole, per ESDU75034

Those curves can be approximated by the following polynomials:

Shear buckling correction: $K/K_0 = 0.95 - 0.96 \cdot (d/b)$

Stress concentration: $f_{max}/q = 4 + 37 \cdot (d/b)^3$

The first step shall be to calculate the buckling loads of the web as if no hole were present using methods proposed in chapter 3450 "PLATE/SHELL BUCKLING".

The correction factor proposed by ESDU75034 shall be then applied to account the effect of the hole.

The post-buckling behavior of the holed plate shall not be discussed here.

Provided no further detailed numerical analysis, validated by dedicated tests were available, the recommended strategy to size this kind of parts shall be to use simplified methods above to prevent buckling up to ultimate loads and to include stiffening features to the design as folded lips at hole edge.

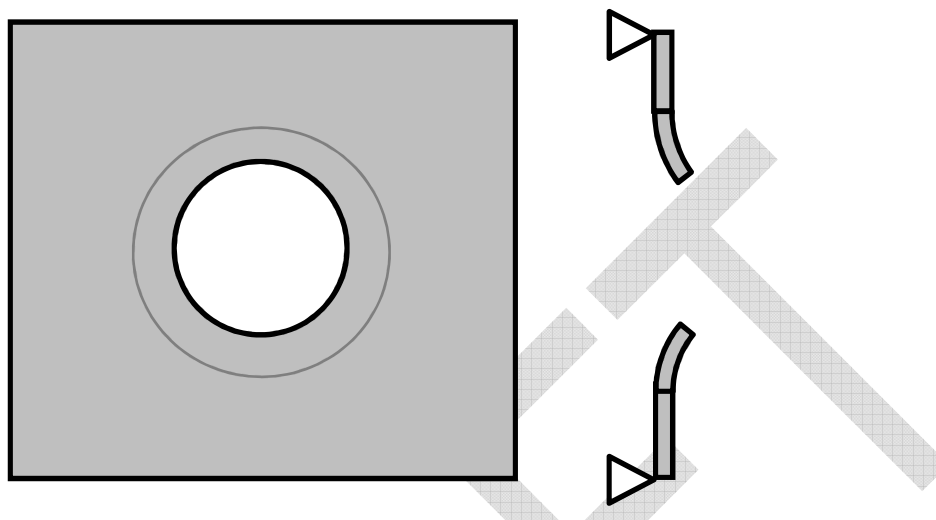


Figure 4. Folded stiffening lips at hole edge to retard buckling onset

3 METHODS FOR FINAL STRENGTH CHECK

The methods presented in previous section to perform the calculation of holed composite plates are just rough approximation and shall not be used for final sizing steps or certification justification.

Unless further validations are done for these approximated methodologies, dedicated FEM shall be used to complete the structural justification of this type of parts.

4 APPENDICES

ABBREVIATIONS AND SYMBOLS

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
[A]	N/mm	In-plane stiffness matrix of the composite laminate
[ABBD]	(*)	Condensed notation for the 6x6 matrix composed by [A], [B] and [D] in Classical Laminate Theory
b	mm	Width of the plate
[B]	N	Bending-extension coupling stiffness matrix of the composite laminate
BVID	-	Barely visual impact damage
CAI	-	Residual compression strength of a laminate including a BVID
d	mm	Diameter of the hole
[D]	Nmm	Flexural stiffness matrix of a composite laminate
ϵ	-	Strain at a plane within the laminate
κ	-	Curvature strain at the mid-surface of the laminate
K_T	-	Stress concentration factor
K_{T0}	-	Stress concentration factor of the infinite plate
M	N	Bending load in the laminate plate
N	N/mm	In-plane load in the laminate
t	mm	Thickness of the laminate
TAI	-	Residual tension strength of a laminate including a BVID
x,y,z	-	As subscripts, direction in the local laminate reference system, with X aligned with material stacking reference (typically with plies at 0°), and Z through thickness direction

(*) Units of that [ABBD] matrix are different for each term (see units for [A], for [B] and for [D] terms).

REFERENCES

	<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1	[S. G. Lekhnitskii]	1984	Anisotropic Plates
Ref. 2	[SAE] CMH-17	2012	Composite Materials Handbook (Volume 3)
Ref. 3	ESDU85001	-	ELASTIC STRESS AND STRAIN DISTRIBUTIONS AROUND CIRCULAR HOLES IN INFINITE PLATES OF ORTHOTROPIC MATERIAL
Ref. 4	ESDU75034	-	INITIAL BUCKLING STRESS, MAXIMUM DIRECT STRESS AND SHEAR STRAIN OF

HOLED PLATES AND WEBS

SQUARE PLATES IN SHEAR WITH CENTRAL CIRCULAR HOLES

Ref. 5 [S.C. Tan]

1988 Finite-Width Correction Factors for Anisotropic Plate Containing a Central Opening

SOFTWARE

<i>Program</i>	<i>Version</i>	<i>Description</i>
STRENGTH2000 (Airbus DS)	TBC	Detailed sizing of metallic and composite aircraft structures
NASTRAN (MSC-Software)	2005	Multidisciplinary structural analysis software based in the Finite Elements method

Table 1. *Software programs referenced in this chapter*

DISTRIBUTION LIST

<i>Department</i>	<i>Name</i>	<i>Full Doc.</i>	<i>Cover Page</i>
Stress Methods & Optimisation (TEASP-TL3)	Fernass Daoud		X
Fatigue & Damage Tolerance GET (TEASA-TL1)	Javier Gomez-Escalonilla Martin		
Stress M&L, Derivatives, Mission (TEASA-TL2)	Miguel Angel Calero Gallardo	X	
Stress Combat MAN (TEASA-TL3)	Sebastian Rönz	X	
Stress Combat, Aerostructures, A400M and Technologies GET (TEASA-TL4)	Sofia Ponce Borrero	X	
Structural Analysis BRE (TEASA-TL5)	Christian Frank		
Structural Repair GET (TEASA-TL6)	Jose Luis Melchor Martin de la Sierra	X	
Fatigue & Damage Tolerance MAN (TEASA-TL7)	Patrick Schneider		

Table 1. *Distribution list (of electronic document, ahead of public release)*

HOLED PLATES AND WEBS

RECORD OF REVISIONS

Revision Date	Authors	Change Reason	Pages/Sub-chapters affected
0.3 29/11/2018	G. Baños	First issue	All pages

Table 2. Record of Revisions: authors, change reasons and sections/pages affected

APPROVAL SIGNATURES TABLE

Dpt. \ Site	Getafe	Manching
Stress Analysis (TEASA)	Miguel Ángel Calero XX/XX/2018	Bernhard Mayer XX/XX/2018
Stress Methods (TEASP-TL3)	G. Baños 29/11/2018	Florian Glock XX/XX/2018

Table 3. Approval signatures table for last revision of this chapter